# REDUCTION MODULO $p$ AND A BOUND ON THE TORSION

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The aim is a `Magma` script, written from scratch, that bounds the torsion subgroup of an elliptic curve over $\mathbb{Q}$ using only the local theory of [Sil09, §§VII.1–VII.3]. Section 1 shows that reduction modulo an odd prime of good reduction is injective on the rational torsion points, section 2 turns this into two bounds, and section 3 tests them against the LMFDB. Each problem prints a piece of the script with its output. You prove the mathematics behind it. Each problem is preceded by the definitions and results it uses.

**Standing notation.**

- An elliptic curve $E$ over $\mathbb{Q}$, given by a Weierstrass equation with coefficients $a_1, \dots, a_6 \in \mathbb{Z}$ and discriminant $\Delta$.
- The torsion subgroup $T \coloneqq E(\mathbb{Q})_{\mathrm{tors}}$.
- A prime $p$ is good if $p \nmid \Delta$. Then reducing the equation modulo $p$ gives an elliptic curve $E_p$ over $\mathbb{F}_p$.
- For an abelian group $G$, $G[m]$ is the subgroup killed by $m$.
- A finite abelian group $G$ is isomorphic to $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$ for at most one pair of integers $1 \leqslant n_1 \mid n_2$. When it is, we write $n_1(G)$ and $n_2(G)$ for them.

**How to use this.** Allow 20 minutes for §1 and 30 for §2. Section 3 is for home. Every `Magma` block runs in seconds in the free online calculator. Paste them in order, since later blocks use earlier functions.

## 1. Reduction is injective on torsion

Everything in this section is in [Sil09, §§VII.1–VII.3]. The curve of Problem 1.6 is from Application VII.3.5 there.

In the following definition and results, $p$ is a good prime for $E$.

**Definition 1.1** (Reduction modulo $p$)**.** We regard $E(\mathbb{Q})$ as a subgroup of $E(\mathbb{Q}_p)$, and write $P \mapsto \bar{P}$ for the reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ and $E_1(\mathbb{Q}_p)$ for its kernel.

**Proposition 1.2** (Minimality [Sil09, Remark VII.1.1])**.** *The equation is minimal at $p$, because $v_p(\Delta) = 0 < 12$. So $E_p$ is the reduction of $E$ modulo $p$.*

**Proposition 1.3** (Reduction is a homomorphism [Sil09, Proposition VII.2.1 and the discussion after it])**.** *The reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ is a surjective group homomorphism. Its kernel $E_1(\mathbb{Q}_p)$ consists of $O$ and the $(x, y) \in E(\mathbb{Q}_p)$ with $v_p(x) < 0$.*

**Proposition 1.4** (Torsion prime to $p$ [Sil09, Proposition VII.3.1])**.** *If $p \nmid m$, then $E_1(\mathbb{Q}_p)[m] = \{O\}$, and the reduction map above restricts to an injection $E(\mathbb{Q}_p)[m] \hookrightarrow E_p(\mathbb{F}_p)$.*

**Theorem 1.5** (Torsion of $p$-power order [Sil09, Theorem VII.3.4(b)])**.** *Let $P \in E(\mathbb{Q}_p)$ have exact order $p^n$ with $n \geqslant 1$, and let $r \coloneqq \lfloor 1/(p^n - p^{n-1}) \rfloor$. Then $p^{2r}x(P) \in \mathbb{Z}_p$. In particular $x(P) \in \mathbb{Z}_p$ unless $p = 2$ and $n = 1$.*

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**Problem 1.6** (Injectivity of reduction on torsion)**.**
(1) Let $p \geqslant 3$ be a good prime. Show that the reduction map

$$E(\mathbb{Q}) \hookrightarrow E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$$

is injective on $T$. In particular $T$ is finite, and isomorphic to a subgroup of $E_p(\mathbb{F}_p)$.[a]
(2) The prime 2 is different: even if $p = 2$ is good, $T \to E_2(\mathbb{F}_2)$ need not be injective. Take for instance the curve $E\colon y^2 + xy = x^3 + 4x + 1$:

```
E := EllipticCurve([1, 0, 0, 4, 1]);          // y^2 + xy = x^3 + 4x + 1
Factorization(Integers()!Discriminant(E));    // [ <5, 2>, <13, 2> ]
P := E![-1/4, 1/8];  2*P eq E!0;              // true
```

So 2 is a good prime and $P = (-1/4, 1/8)$ is a rational point of order 2. Show that $\bar{P} = O$ in $E_2(\mathbb{F}_2)$. Which hypothesis of Proposition 1.4 and Theorem 1.5 fails, and what does Proposition 1.4 still give at $p = 2$?

[a] *Hint:* a nonzero torsion point in $E_1(\mathbb{Q}_p)$ has a multiple of prime order in $E_1(\mathbb{Q}_p)$.

## 2. Two bounds

This section is [Sil09, Application VII.3.2], and its three curves are [Sil09, Examples VII.3.3.1–VII.3.3.3].

**Problem 2.1** (The order bound)**.**
The first piece of the script computes the greatest common divisor $b$ of the orders $\#E_p(\mathbb{F}_p)$ over the good odd primes $p \leqslant N$:

```
// b = gcd of #E_p(F_p) over the good odd primes p <= N (there must be one)
OrderBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  return GCD([#ChangeRing(E, GF(p)) : p in good]);
end function;
E1 := EllipticCurve([0, 0, 1, -1, 1]);          // y^2 + y = x^3 - x + 1
Discriminant(E1);                               // -611
[#ChangeRing(E1, GF(p)) : p in [3, 5, 7]];      // [ 1, 8, 6 ]
OrderBound(E1, 30);                             // 1
E2 := EllipticCurve([0, 3]);                    // y^2 = x^3 + 3
Discriminant(E2);                               // -3888
[#ChangeRing(E2, GF(p)) : p in [5, 7, 11, 13]]; // [ 6, 13, 12, 9 ]
OrderBound(E2, 30);                             // 1
```

(1) Deduce that both curves have trivial torsion, and that $y^2 = x^3 + 3$ has infinitely many rational points.
(2) `OrderBound` skips the primes dividing the discriminant of the equation it is given. Compare two equations for the first curve, with $N = 7$:

```
E4 := EllipticCurve([0, 0, 27, -81, 729]);      // y^2 + 27y = x^3 - 81x + 729
Discriminant(E1), Discriminant(E4);             // -611 -324710451
IsIsomorphic(E1, E4);                           // true
```

```
OrderBound(E1, 7), OrderBound(E4, 7);            // 1 2
```

Here $-324710451 = -3^{12} \cdot 611$. Find a change of variables between the two equations. Explain why a non-minimal equation can make the script skip a prime it could have used, but never makes it use one it should not.

**Corollary 2.2** (The $m$-torsion [Sil09, Corollary III.6.4]). *Let $E$ be an elliptic curve over a field $K$, and $m \geqslant 1$. If $\operatorname{char} K \nmid m$, then $E[m](\overline{K}) \cong (\mathbb{Z}/m\mathbb{Z})^2$. If $\operatorname{char} K = p > 0$, then $E[p^e](\overline{K})$ is cyclic for every $e \geqslant 1$.*

**Problem 2.3** (The structure bound).
The curve $y^2 = x^3 + x$ shows that the orders are not the whole story. `Invariants` returns $[n_1(G), n_2(G)]$ for $G = E_p(\mathbb{F}_p)$, omitting $n_1(G)$ when it is 1, and returns $[\,]$ when $G$ is trivial:

```
E3 := EllipticCurve([1, 0]);                     // y^2 = x^3 + x
Discriminant(E3);                                // -64
OrderBound(E3, 30);                              // 4
[Invariants(AbelianGroup(ChangeRing(E3, GF(p)))) : p in [3, 5, 7, 11, 13]];
// [ [ 4 ], [ 2, 2 ], [ 8 ], [ 12 ], [ 2, 10 ] ]
```

(1) Let $G$ be a finite subgroup of $E(\overline{K})$, for an elliptic curve $E$ over a field $K$. Show that $n_1(G)$ and $n_2(G)$ are defined, and that $n_2(G)$ is the exponent of $G$. Show that $\#E[n_1(G)](\overline{K}) = n_1(G)^2$, and that an integer $m \geqslant 1$ with $\#E[m](\overline{K}) = m^2$ satisfies $E[m](\overline{K}) \subseteq G$ if and only if $m \mid n_1(G)$. Deduce that $n_1(H) \mid n_1(G)$ and $n_2(H) \mid n_2(G)$ for every subgroup $H \subseteq G$.[a]
(2) Let $b_1$ and $b_2$ be the greatest common divisors of $n_1(E_p(\mathbb{F}_p))$ and of $n_2(E_p(\mathbb{F}_p))$ over the good odd primes $p \leqslant N$. Show that $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $b_1 b_2 \mid b$. This is the second piece of the script:

```
// [b1, b2]: n1(T) | b1 and n2(T) | b2
StructureBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  b1 := 0;  b2 := 0;
  for p in good do
    I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
    b1 := GCD(b1, #I eq 2 select I[1] else 1);
    b2 := GCD(b2, #I eq 0 select 1 else I[#I]);
  end for;
  return [b1, b2];
end function;
StructureBound(E3, 30);                          // [ 1, 2 ]
P := E3![0, 0];  2*P eq E3!0;                    // true
```

Conclude that $T = \{O, (0,0)\} \cong \mathbb{Z}/2\mathbb{Z}$ for $y^2 = x^3 + x$. Why does $b$ stay at 4 however large $N$ is?
(3) Show that a rational point $P$ is a torsion point if and only if $b_2 P = O$, and that rational torsion points generating a subgroup of order $b_1 b_2$ generate $T$.

[a] *Hint:* for a prime $\ell$, the group $G[\ell]$ lies in $E[\ell](\overline{K})$. Use Corollary 2.2.

## 3. How good are the bounds? The LMFDB

The L-functions and modular forms database [LMF25] lists millions of elliptic curves over $\mathbb{Q}$ with their torsion subgroups, computed by methods beyond [Sil09, Chapter VII]. We use it as the answer key.

**Proposition 3.1** (Reduction of isogenies [Mil20, proof of Theorem V.4.1])**.** *An isogeny over $\mathbb{Q}$ reduces to an isogeny over $\mathbb{F}_p$ at every good prime $p$ of both equations.*

**Proposition 3.2** (Isogenous curves over $\mathbb{F}_p$ [Sil09, Exercise 5.4(a)])**.** *Isogenous curves over $\mathbb{F}_p$ have the same number of rational points.*

**Problem 3.3** (Testing the bounds)**.**
(1) Here are the curves of smallest conductor with each torsion structure that occurs over $\mathbb{Q}$, with their LMFDB labels, $a$-invariants and torsion structures ([1] for the trivial group):

```
curves := [<"11.a1", [0, -1, 1, -7820, -263580], [1]>,
  <"14.a2", [1, 0, 1, -171, -874], [2]>,
  <"19.a2", [0, 1, 1, -9, -15], [3]>, <"15.a4", [1, 1, 1, -80, 242], [4]>,
  <"11.a2", [0, -1, 1, -10, -20], [5]>, <"14.a6", [1, 0, 1, 4, -6], [6]>,
  <"26.b2", [1, -1, 1, -3, 3], [7]>, <"15.a8", [1, 1, 1, 35, -28], [8]>,
  <"54.b2", [1, -1, 1, -14, 29], [9]>, <"66.c3", [1, 0, 0, -45, 81], [10]>,
  <"90.c7", [1, -1, 1, -122, 1721], [12]>, <"15.a2", [1, 1, 1, -135, -660], [2, 2]>,
  <"15.a5", [1, 1, 1, -10, -10], [2, 4]>, <"30.a6", [1, 0, 1, -19, 26], [2, 6]>,
  <"210.e6", [1, 0, 0, -1070, 7812], [2, 8]>];
for c in curves do
  E := EllipticCurve(c[2]);
  printf "%-7o LMFDB %-9o order bound %-3o structure bound %o\n",
    c[1], c[3], OrderBound(E, 50), StructureBound(E, 50);
end for;
// 11.a1   LMFDB [ 1 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a2   LMFDB [ 2 ]     order bound 6   structure bound [ 1, 6 ]
// 19.a2   LMFDB [ 3 ]     order bound 3   structure bound [ 1, 3 ]
// 15.a4   LMFDB [ 4 ]     order bound 8   structure bound [ 1, 4 ]
// 11.a2   LMFDB [ 5 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a6   LMFDB [ 6 ]     order bound 6   structure bound [ 1, 6 ]
// 26.b2   LMFDB [ 7 ]     order bound 7   structure bound [ 1, 7 ]
// 15.a8   LMFDB [ 8 ]     order bound 8   structure bound [ 1, 8 ]
// 54.b2   LMFDB [ 9 ]     order bound 9   structure bound [ 1, 9 ]
// 66.c3   LMFDB [ 10 ]    order bound 10  structure bound [ 1, 10 ]
// 90.c7   LMFDB [ 12 ]    order bound 12  structure bound [ 1, 12 ]
// 15.a2   LMFDB [ 2, 2 ]  order bound 8   structure bound [ 2, 4 ]
// 15.a5   LMFDB [ 2, 4 ]  order bound 8   structure bound [ 2, 4 ]
// 30.a6   LMFDB [ 2, 6 ]  order bound 12  structure bound [ 2, 6 ]
// 210.e6  LMFDB [ 2, 8 ]  order bound 16  structure bound [ 2, 8 ]
```

Check that no bound contradicts the LMFDB. Where is each bound attained?
(2) The curves 15.a4 and 15.a8 are isogenous:

```
E := EllipticCurve([1, 1, 1, -80, 242]);        // 15.a4
F := EllipticCurve([1, 1, 1, 35, -28]);          // 15.a8
good := [p : p in PrimesUpTo(50) | p ne 3 and p ne 5];
[#ChangeRing(E, GF(p)) : p in good] eq [#ChangeRing(F, GF(p)) : p in good];
// true
```

```
[<p, Invariants(AbelianGroup(ChangeRing(E, GF(p)))),
     Invariants(AbelianGroup(ChangeRing(F, GF(p))))> : p in [7, 11, 13]];
// [ <7, [ 2, 4 ], [ 8 ]>, <11, [ 2, 8 ], [ 16 ]>, <13, [ 16 ], [ 2, 8 ]> ]
```

Using Propositions 3.1 and 3.2, explain why the order bound cannot separate isogenous curves, and why the structure bound separates these two.
(3) The LMFDB lists $(5, -2)$ as a torsion generator of 15.a4:

```
E := EllipticCurve([1, 1, 1, -80, 242]);  P := E![5, -2];        // 15.a4
[k*P eq E!0 : k in [1..4]];                        // [ false, false, false, true ]
```

Prove that $T \cong \mathbb{Z}/4\mathbb{Z}$ for 15.a4 without trusting the LMFDB. Can you do the same for 11.a1, 14.a2 or 15.a2?
(4) Find these curves yourself: on https://www.lmfdb.org/EllipticCurve/Q/, search by "torsion structure", such as `[2,4]`, and sort by conductor. On a curve's page, find its equation, discriminant, torsion subgroup and bad primes. Pick a curve of your own and compute both bounds.

**Remark 3.4** (Beyond the local theory)**.** Over $\mathbb{Q}$ the bounds complete to an algorithm. For $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$, a rational torsion point $P \neq O$ has integer coordinates, and either $2P = O$ or $y(P)^2 \mid 4A^3 + 27B^2$ (Nagell–Lutz, [Sil09, Corollary VIII.7.2]). This leaves finitely many candidates, and the test $b_2 P = O$ decides which are torsion points. Mazur proved that $E(\mathbb{Q})_{\text{tors}}$ is one of the fifteen groups in the listing [Sil09, Theorem VIII.7.5].

## References


[LMF25] The LMFDB Collaboration, *The L-functions and modular forms database*, 2025, https://www.lmfdb.org. [Online; accessed 21 July 2025]. ↑

[Mil20] James S. Milne, *Elliptic curves*, Second, World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2020. ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑